Beyond Item Popularity: Modelling Repetitive Consumption with Quantised Item Representations*

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Abstract

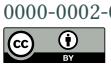
In music recommendation, sequential recommenders must account for repetitive listening, as users frequently replay familiar tracks. Personalised Popularity Scores (PPS) capture these patterns from user-specific historical frequencies, improving relevance but often reinforcing already-known content at the expense of novel or serendipitous recommendations. To address this limitation, we build upon RecJPQ, a Transformer-based framework based on sub-item decomposition, and repurpose its architecture to model personalised popularity at a finer granularity. We introduce sub-ID-level Personalised Popularity Scores (sPPS), which capture shared repetition patterns across sub-embeddings and enable explicit control over the trade-off between accuracy and personalised novelty. Experiments show that sPPS achieves significantly higher personalised novelty than item-level PPS while preserving recommendation accuracy.

Keywords

Recommender Systems, Sequential Recommendation, Personalised Popularity, Product Quantisation

1. Introduction

Sequential recommendation predicts future user interests from temporally ordered interactions, unlike traditional collaborative filtering approaches that treat histories as unordered sets [2, 3]. Recent advances largely rely on Transformer-based architectures, with representative models including SASRec [4], gSASRec [5], and BERT4Rec [6]. However, music recommendation remains challenging because consumption is highly repetitive and often passive, catalogues are large, and user histories sparse [7, 8]. Popularity provides an effective signal for repetitive consumption [9, 10, 11]. Personalised Popularity Scores (PPS) [12] exploit user-specific recurrence to improve prediction in repetition-heavy settings. However, reinforcing frequently consumed items can limit exposure to novel content and reduce serendipity [13, 14, 15, 16]. To address this limitation, we build on RecJPQ [17], which represents items as combinations of shared sub-embeddings through sub-item decomposition. Related sub-ID representations have shown strong performance and generalisation [18, 19]. We repurpose this architecture to model personalised popularity at the sub-ID level and compare it with item-level popularity under an accuracy–novelty trade-off. Our contributions are twofold: (1) we introduce personalised popularity modelling at both item and sub-ID levels; and (2) we evaluate their effects on sequential music recommendation. Results show that sub-ID popularity captures repetition and similarities across items while enabling explicit control over the accuracy–novelty trade-off.

Proceedings of the 16th Italian Information Retrieval Workshop, IIR 2026, November 12–14, 2026, Rome, Italy
*Extended abstract. The full paper is available in the RecSys 2025 Proceedings [1].
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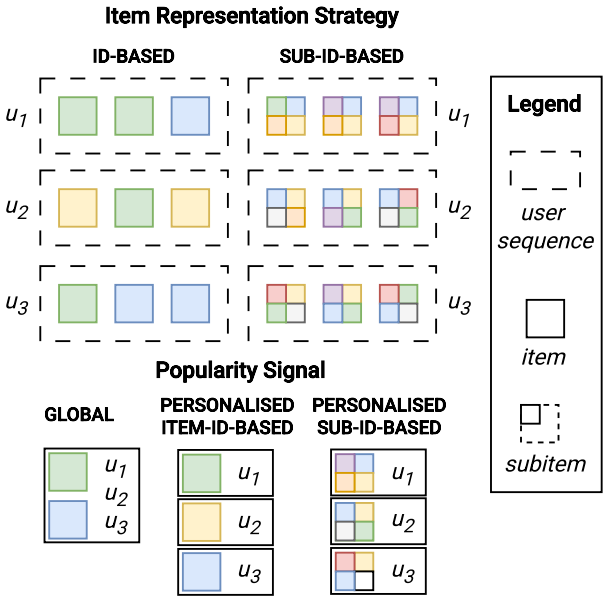


Figure 1: Different approaches for computing item popularity, based on whether the signal is ID-based (global or personalised) or sub-ID-based (personalised).

2. Methodology

We integrate two user-specific popularity signals: item-level Personalised Popularity Scores (PPS) and our proposed sub-ID-level scores (sPPS). Figure 1 illustrates both.

Our method builds on RecJPQ [17], which applies Product Quantisation (PQ) [20] and Joint Product Quantisation (JPQ) [21] to sequential recommendation. PQ partitions a d-dimensional item embedding into m segments mapped to a codebook of size V, while JPQ enables end-to-end learning of the sub-embeddings. RecJPQ assigns sub-IDs through SVD decomposition of the sequence–item interaction matrix, representing each item i as:

$$\text{code}(i) = [z_1^{(i)}, \ldots, z_m^{(i)}], \qquad z_k^{(i)} \in \{0, \ldots, V-1\}. \tag{1}$$

Each $z_k^{(i)}$ identifies a sub-embedding, and the item embedding is reconstructed by concatenating the m corresponding sub-embeddings [22, 23].

2.1. Modelling Personalised Popularity at the Item Level

PPS [12] captures explicit repetition by measuring how frequently each item occurs in a user's interaction sequence $\mathcal{S}_u$. Given the count c_i of item i, its score is:

$$\text{PPS}_i = \log(c_i + \epsilon). \tag{2}$$

The logarithmic transformation smooths repetition counts, while Z-score normalisation makes scores comparable:

$$\text{PPS}_i^{\text{std}} = \frac{\text{PPS}_i - \mu_{\text{PPS}}}{\sigma_{\text{PPS}}}, \tag{3}$$

where μ_{PPS} and σ_{PPS} are the mean and standard deviation over the item set.

2.2. Modelling Personalised Popularity at the Sub-ID Level

We extend personalised popularity from item IDs to RecJPQ sub-IDs. Since SVD-based quantisation associates shared sub-IDs with latent item characteristics, their recurrence can capture preferences across items beyond exact-item repetition. For each split $j \in \{1, \ldots, m\}$, we count sub-ID k in user u's

Table 1
Statistics of the two music datasets after popularity-based sampling of 30,000 items. *Avg. len* and *Med. len* denote the average and median interactions per user.

Dataset	Users	Items	Interactions	Avg. len	Med. len
Yandex	20862	30000	28408152	1362	1275
Last.fm-1K	990	30000	4990042	5040	2605

history as $c_j^{(u)}[k] = \sum_{i \in \mathcal{S}_u} \mathbf{1}\{z_j^{(i)} = k\}$. For candidate item i, we aggregate the counts of its m sub-IDs and apply logarithmic smoothing [24] to obtain the ***sub-ID-based Personalised Popularity Score (sPPS)***:

$$\mathrm{sPPS}_i = \sum_{j=1}^{m} \log\left(c_j^{(u)}[z_j^{(i)}] + \epsilon\right). \tag{4}$$

The score is standardised as in Eq. (3), yielding $\mathrm{sPPS}_i^{\mathrm{std}}$.

2.3. Score Function

We combine the recommender output with PPS and sPPS as:

$$\mathrm{logits}_i^{\mathrm{final}} = \gamma \cdot \mathrm{logits}_i^{\mathrm{rec}} + \alpha \cdot \mathrm{PPS}_i^{\mathrm{std}} + \beta \cdot \mathrm{sPPS}_i^{\mathrm{std}}, \tag{5}$$

where α and β control PPS and sPPS, respectively, and $\gamma = 1 - \alpha - \beta$ ensures a convex combination, controlling the trade-off between item-level memorisation, sub-ID generalisation, and the sequential recommender.

3. Experimental Setup

Research Questions. We investigate two research questions: **RQ1**, how integrating item-level personalised popularity (PPS) affects RecJPQ recommendation performance; and **RQ2**, whether sub-ID popularity (sPPS) improves the accuracy–novelty trade-off compared with PPS. Our setup follows [12] in dataset selection and evaluation. Code, data, and replication instructions are publicly available.[1]
Datasets. We evaluate on Yandex Music[2] and Last.fm-1K[3] [7], using the preprocessing protocol of [12]. Table 1 reports the main statistics.
Model. We evaluate BERT4Rec [6] with RecJPQ [17], integrated into the aprec[4] reproducibility framework. Based on preliminary experiments and consistently with RecJPQ, we use embedding dimension $d = 256$ and $m = 32$ codebook splits. We fix $\alpha = 0.4$ and $\beta = 0.4$ during training and systematically vary them at inference to analyse different accuracy–novelty trade-offs without retraining.
Evaluation Details. Following [12], we apply a Global Temporal Split, sorting interactions by timestamp and reserving the last 10% for testing. Validation uses 10% of users (2,048 for Yandex and 128 for Last.fm-1K). Accuracy is measured using NDCG [25]. For Yandex, following Yakura et al. [26, 27], relevance is mapped as *like*=2, *play*=1, *skip*=-1, and *dislike*=-2. For repeated interactions with a track, the label with the greatest absolute value is retained, and negative labels are treated as zero relevance [28]. Personalised novelty follows [29, 30]:

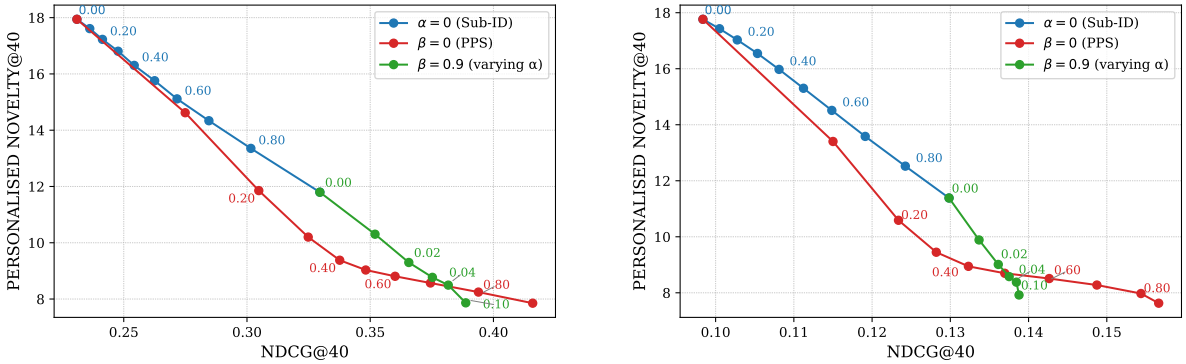
$$\mathrm{Novelty@K}_u = \frac{1}{K} \sum_{i \in R_u@K} -\log_2 p(i \mid u), \tag{6}$$

where $R_u@K$ is the recommendation list and $p(i \mid u) = \max(c_i / \sum_j c_j, 10^{-8})$ is estimated from the user's interaction history. Thus, frequent items receive lower novelty. All evaluations use $K = 40$.

[1] https://github.com/sisinflab/Sub-id-Popularity
[2] https://www.kaggle.com/competitions/yandex-music-event-2019-02-16
[3] http://ocelma.net/MusicRecommendationDataset/lastfm-1K.html
[4] https://github.com/asash/BERT4Rec_repro

Table 2
Comparison of NDCG@40 at increasing novelty thresholds for PPS and sPPS on Last.fm and Yandex.

Dataset	Method	Novelty ≥ 0	Novelty ≥ 10	Novelty ≥ 12	Novelty ≥ 14
Last.fm	PPS	**0.4159**	0.3248	0.2749	0.2749
	sPPS	0.3296	**0.3296**	**0.3016**	**0.2846**
Yandex	PPS	**0.1566**	0.1233	0.1150	0.0983
	sPPS	0.1298	**0.1298**	**0.1242**	**0.1149**



(a) Last.fm (b) Yandex

Figure 2: Accuracy–novelty trade-off for PPS (red), sPPS (blue), and their combination (green), with $\beta = 0.9$ fixed while increasing α.

4. Results

Table 2 and Figure 2 summarise the accuracy–novelty trade-off of item- and sub-ID popularity.
RQ1. Varying α while keeping $\beta = 0$ shows that increasing PPS consistently improves NDCG on both datasets (Figure 2, red line), confirming the benefit of modelling exact-item recurrence. PPS also achieves higher maximum NDCG than sPPS when novelty is unconstrained (Table 2). However, this comes at the cost of personalised novelty: on Last.fm, it decreases from 14.62 at $\alpha = 0.1$ to 7.85 at $\alpha = 0.9$, with a similar trend on Yandex. Thus, item-level popularity improves predictive accuracy while increasingly favouring familiar content.
RQ2. Fixing $\alpha = 0$ and varying β isolates the sub-ID signal. sPPS consistently provides higher personalised novelty than PPS (Figure 2, blue line). On Last.fm, at NDCG@40 ≈ 0.32, sPPS reaches novelty ≈ 12 versus ≈ 10 for PPS, a 20% relative improvement at comparable accuracy. At a novelty threshold of 12, NDCG@40 increases from 0.2749 to 0.3016 (+9.7%) on Last.fm and from 0.1150 to 0.1242 (+8.0%) on Yandex. This indicates that sub-ID recurrence captures preferences beyond exact-item repetition, improving novelty without sacrificing comparable accuracy. Finally, combining both signals by fixing $\beta = 0.9$ and increasing α steadily improves NDCG while reducing novelty less than PPS alone (green curves in Figure 2). This suggests that sPPS provides a novelty-preserving foundation on which item-level PPS can recover additional accuracy.

5. Conclusion

This work shows that personalised popularity can improve sequential music recommendation. Extending RecJPQ with item- and sub-ID-level popularity enables fine-grained personalisation while preserving sub-item decomposition. Experiments on Yandex and Last.fm-1K show that sPPS improves personalised novelty at comparable accuracy, capturing recurrence beyond exact-item repetition and providing control over the accuracy–novelty trade-off. Future work will investigate different α and β configurations during training to better understand their interaction across user behaviour profiles.

Declaration on Generative AI

During preparation of this work, the authors used Generative AI for grammar and spelling checks. The authors reviewed and edited the content and take full responsibility for the publication.

Acknowledgments

The authors acknowledge partial support of OVS Fashion Retail Reloaded, Lutech Digitale 4.0, Servizi Locali 2.0, Konecta New KOCS ERA, ACSOC, DIRO-SAT, RAIL-NEXT and Konecta 2-AURA. We acknowledge the CINECA award under the ISCRA initiative for access to the LEONARDO supercomputer, owned by the EuroHPC Joint Undertaking and hosted by CINECA (Italy). The authors would also like to thank Aleksandr V. Petrov for his valuable feedback and insightful discussions that contributed to this work.

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